2. Given the following transaction data set, please answer the questions below:

A database has five transactions. Let min_sup=60%, min_sup=60%, and min_conf=80%.

TID	Items
$T100$	{M,O,N,K,E,Y}
$T200$	{D,O,N,K,E,Y}
$T300$	{M,A,K,E}
$T400$	{M,U,C,K,Y}
$T500$	{C,O,O,K,I,E}

1. Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes. (24 mins)
2. List all of the strong association rules, satisfying the minimum support and confidence thresholds (6 min)

Answer:

(a) Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.

Apriori:

C1 =

m	3
o	3
n	2
k	5
e	4
y	3
d	1
a	1
u	1
c	2
i	1

L1 =

m	3
o	3
k	5
e	4
y	3

C2 =

mo	1
mk	3
me	2
my	2
ok	3
oe	3
oy	2
ke	4
ky	3
ey	2

L2 =

mk	3
ok	3
oe	3
ke	4
ky	3

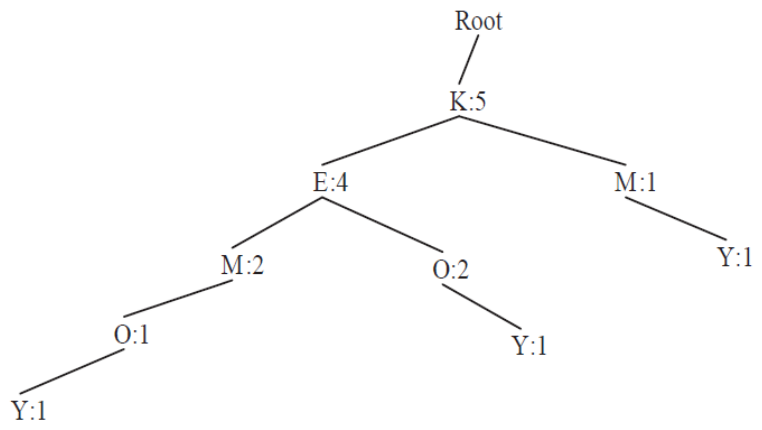
C3=

oke	3
~~key~~	~~2~~

L3=

oke	3

FP-growth:



FP-Tree Table:

item	conditional pattern base	conditional tree	frequent pattern
y	{ {k,e,m,o:1}, {k,e,o:1}, {k,m:1} }	k:3	{k,y:3}
o	{ {k,e,m:1}, {k,e:2}	k:3,e:3	{k,o:3}, {e,o:3}, {k,e,o:3}
m	{ {k,e:2}, {k:1} }	k:3	{k,m: 3}
e	{ {k:4} }	k:4	{ k,e:4 }

Efficiency comparison: Apriori has to do multiple scans of the database while FP-growth builds the FP-Tree with a single scan. Candidate generation in Apriori is expensive (owing to the self-join), while FP-growth does not generate any candidates.

(b) List all of the strong association rules (with support s and confidence c) matching the following metarule, where X is a variable representing customers, and $item_i$ denotes variables representing items (e.g., “A”, “B”, etc.):

Binary partitioning, check confidence

$$\forall x \in transaction,\ buys(X, item_1) \wedge buys(X, item_2) \Rightarrow buys(X, item_3) \quad [s, c]$$

k,o → e [0.6,1]
e,o → k [0.6,1]